

(T3): marginalisation of the log-Sobolev inequality to sharp
time
and which Dirichlet form the programme actually needs

Abstract

(T3) is settled. If the space-cutoff Euclidean measure ν_g on $\mathcal{S}'(\mathbb{R}^2)$ satisfies a Poincaré or log-Sobolev inequality with respect to the *covariance-weighted* Dirichlet form $\mathbb{E}_\nu \langle \nabla F, G \nabla F \rangle$, then its sharp-time marginal μ_g satisfies the same inequality, *with the same constant*, with respect to $\mathbb{E}_\mu \langle \nabla f, C \nabla f \rangle$, $C = (2\mu)^{-1}$ (Theorem 3.2). Two exact facts drive it: the sharp-time restriction of $G = (-\Delta_2 + m_0^2)^{-1}$ is C (Lemma 2.1), and time-mollification only *decreases* the Dirichlet form, so the inequality survives the limit in the favourable direction (Lemma 3.1). Both verified numerically (§5).

A sharper question for (T1). The main theorem of `h6-attack.tex` requires the *covariance-weighted* form. The plain L^2 Dirichlet form is **useless** for it: for a Wick monomial of degree $n \geq 2$, $\mathbb{E}[|\nabla F_n[u]|_{L^2}^2] = +\infty$ identically, because $\mathbb{E}[(\varphi^m(x))^2] = m! C(0)^m = \infty$ (Proposition 4.1). Covariance-weighted LSI implies L^2 -LSI but not conversely. So (T1) must verify not merely *that* Bauerschmidt–Dagallier prove an LSI, but *in which form*. That is now the sharpest single question in the programme.

Update (8 August 2026)

The conditional marginalisation theorem below remains a statement about a *covariance-weighted* two-dimensional inequality. The question posed in the abstract has now been answered by checking [1, Eq. (1.6) and Thm. 1.1]: their LSI uses the unweighted lattice ℓ^2 gradient. It therefore does not satisfy hypothesis (P-2d) and does not feed Theorem 3.2. No comparison reverses this mismatch because the Fourier multiplier of $C = (2\sqrt{-\partial_x^2 + m_0^2})^{-1}$ tends to zero at high momentum. Thus (T3) is a valid conditional lemma, but the proposed $\text{BD} \Rightarrow (\text{P-2d}) \Rightarrow (\text{P-U})$ route is closed. See `remaining.tex` for the corrected program.

1 Statement

Let $G := (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$, Φ_G the centred Gaussian measure on $\mathcal{S}'(\mathbb{R}^2)$ with covariance G , and $d\nu_g = Z_g^{-1} e^{-U_g} d\Phi_G$ with $U_g = \lambda \int_{\mathbb{R}^2} :P(\varphi(t, x)): g(x) dt dx$ the space-cutoff Euclidean measure; by Feynman–Kac–Nelson its sharp-time marginal under $\pi : \varphi \mapsto \varphi(0, \cdot)$ is the ground-state measure $\mu_g = \pi_* \nu_g = \Omega_g^2 d\varphi_0$. Write $\mu = (-\partial_x^2 + m_0^2)^{1/2}$ and $C = (2\mu)^{-1}$.

Definition 1.1 (covariance-weighted Dirichlet form). For a centred Gaussian reference with covariance $\mathcal{C}$ and $\nu \ll \Phi_{\mathcal{C}}$, put $\mathcal{E}_{\mathcal{C}}^\nu(F) := \mathbb{E}_\nu \langle \nabla F, \mathcal{C} \nabla F \rangle$, ∇ the Cameron–Martin gradient. This is the form for which the Gaussian itself satisfies Poincaré with constant one, $\text{Var}_{\Phi_{\mathcal{C}}}(F) \leq \mathbb{E}_{\Phi_{\mathcal{C}}} \langle \nabla F, \mathcal{C} \nabla F \rangle$.

Hypothesis 1.2 ((P-2d)). There is $c_P < \infty$, independent of g , such that, for all F in the form domain,

$$\text{Var}_{\nu_g}(F) \leq c_P \mathbb{E}_{\nu_g} \langle \nabla F, G \nabla F \rangle.$$

2 The sharp-time covariance identity

Lemma 2.1. *For every $k \in \mathbb{R}$, with $\mu(k) = (k^2 + m_0^2)^{1/2}$,*

$$\int_{\mathbb{R}} \frac{dk_0}{2\pi} \frac{1}{k_0^2 + k^2 + m_0^2} = \frac{1}{2\mu(k)}.$$

Consequently $\pi_\Phi_G = \varphi_0$, the centred Gaussian on $\mathcal{S}'(\mathbb{R})$ with covariance $C = (2\mu)^{-1}$; equivalently $G((0, x), (0, y)) = C(x - y)$.*

Proof. $\int_{\mathbb{R}} dk_0 (k_0^2 + a^2)^{-1} = \pi/a$ with $a = \mu(k) > 0$; divide by 2π . A centred Gaussian is determined by its covariance, and the covariance of $\pi_*\Phi_G$ has Fourier symbol $\int \frac{dk_0}{2\pi} (k_0^2 + \mu(k)^2)^{-1}$. $\square$

3 Marginalisation

Fix a smooth probability density $\rho \geq 0$ on $\mathbb{R}$ with $\int \rho = 1$, and set $\rho_\varepsilon(t) = \varepsilon^{-1}\rho(t/\varepsilon)$ and $\varphi_\varepsilon(x) := \int \rho_\varepsilon(t)\varphi(t, x) dt$. For a bounded smooth cylinder functional f on $\mathcal{S}'(\mathbb{R})$ put $F_\varepsilon(\varphi) := f(\varphi_\varepsilon)$.

Lemma 3.1 (mollification decreases the form). $\langle \nabla F_\varepsilon, G \nabla F_\varepsilon \rangle = \int \frac{dk}{2\pi} |\widehat{v}_\varepsilon(k)|^2 s_\varepsilon(k)$, where $v_\varepsilon := (\nabla f)(\varphi_\varepsilon)$ and

$$s_\varepsilon(k) := \int \frac{dk_0}{2\pi} \frac{|\hat{\rho}(\varepsilon k_0)|^2}{k_0^2 + \mu(k)^2}.$$

Moreover $0 \leq s_\varepsilon(k) \leq \widehat{C}(k) = \frac{1}{2\mu(k)}$ for every $\varepsilon > 0$, and $s_\varepsilon(k) \rightarrow \widehat{C}(k)$ as $\varepsilon \downarrow 0$. Hence

$$\langle \nabla F_\varepsilon, G \nabla F_\varepsilon \rangle \leq \langle v_\varepsilon, C v_\varepsilon \rangle \quad \text{pointwise, for every } \varepsilon > 0. \quad (1)$$

Proof. F_ε depends on φ only through φ_ε , so $\delta F_\varepsilon / \delta \varphi(t, x) = \rho_\varepsilon(t) v_\varepsilon(x)$, i.e. $\nabla F_\varepsilon = \rho_\varepsilon \otimes v_\varepsilon$. Pairing with G in Fourier and using $\widehat{\rho}_\varepsilon(k_0) = \hat{\rho}(\varepsilon k_0)$ gives the displayed formula. Since $\rho \geq 0$ and $\int \rho = 1$ we have $|\hat{\rho}| \leq 1$, whence $s_\varepsilon(k) \leq \int \frac{dk_0}{2\pi} (k_0^2 + \mu^2)^{-1} = \widehat{C}(k)$ by Lemma 2.1; convergence is dominated convergence with the same integrable majorant, using $\hat{\rho}(0) = 1$ and continuity of $\hat{\rho}$. $\square$

Theorem 3.2 ((T3)). *Assume (P-2d). Then for every bounded smooth cylinder functional f on $\mathcal{S}'(\mathbb{R})$,*

$$\text{Var}_{\mu_g}(f) \leq c_P \mathbb{E}_{\mu_g} \langle \nabla f, C \nabla f \rangle,$$

with the same constant c_P , uniformly in g . The same statement holds with Var replaced by $\text{Ent}(\cdot^2)$ and c_P by the log-Sobolev constant.

Proof. Apply (P-2d) to F_ε and use (1):

$$\text{Var}_{\nu_g}(F_\varepsilon) \leq c_P \mathbb{E}_{\nu_g} \langle \nabla F_\varepsilon, G \nabla F_\varepsilon \rangle \leq c_P \mathbb{E}_{\nu_g} \langle v_\varepsilon, C v_\varepsilon \rangle.$$

The sharp-time restriction exists for $P(\varphi)_2$, so $\varphi_\varepsilon \rightarrow \varphi(0, \cdot)$ in ν_g -probability; as f and ∇f are bounded and continuous, bounded convergence gives $\text{Var}_{\nu_g}(F_\varepsilon) \rightarrow \text{Var}_{\nu_g}(f \circ \pi) = \text{Var}_{\mu_g}(f)$ and $\mathbb{E}_{\nu_g} \langle v_\varepsilon, C v_\varepsilon \rangle \rightarrow \mathbb{E}_{\mu_g} \langle \nabla f, C \nabla f \rangle$. Passing to the limit gives the claim. The entropy version is identical, the only change being that $\text{Ent}_{\nu_g}(F_\varepsilon^2) \rightarrow \text{Ent}_{\mu_g}(f^2)$, which holds for f bounded above and below away from zero; the general case follows by the usual truncation. $\square$

Remark 3.3 (why the direction is favourable). The whole content is (1): smearing in time can only *shrink* the Dirichlet form, because $|\hat{\rho}| \leq 1$. Had the inequality gone the other way, the limit $\varepsilon \downarrow 0$ would have destroyed the estimate. No constant is lost.

4 Which Dirichlet form: a sharper question for (T1)

Proposition 4.1. *Let $F_n[u] = \int u : \varphi^n$: with $u \not\equiv 0$ and $n \geq 2$. Then the unweighted Dirichlet form is identically infinite:*

$$\mathbb{E}_{\varphi_0} \left[|\nabla F_n[u]|_{L^2}^2 \right] = n^2 \int u(x)^2 \mathbb{E}[(\varphi(x)^{n-1})^2] dx = n^2 (n-1)! C(0)^{n-1} \int u^2 = +\infty,$$

since $C(0) = +\infty$ logarithmically in one space dimension.

Proof. $\nabla F_n[u](x) = n u(x) : \varphi(x)^{n-1}$., and $\mathbb{E}[(\varphi(x)^m)^2] = m! C(x, x)^m$ by the Wick isometry. $\square$

Remark 4.2 (consequence). The main theorem of `h6-attack.tex` is driven by the identity $\langle \nabla F_n[u], C \nabla F_n[u] \rangle = n^2 \iint u(x) u(y) C(x-y) : \varphi(x)^{n-1} :: \varphi(y)^{n-1}$: the factor $C(x-y)$ supplied by the *weighting* is exactly what removes the coincident-point singularity that Proposition 4.1 exhibits. Therefore:

- a covariance-weighted LSI **is** what the programme needs;
- it implies the L^2 version, since $\langle \nabla F, C \nabla F \rangle \leq \|C\|_{\text{op}} |\nabla F|_{L^2}^2 = (2m_0)^{-1} |\nabla F|_{L^2}^2$;
- the converse fails, because $\widehat{C}(k) = (2\mu(k))^{-1} \rightarrow 0$ as $|k| \rightarrow \infty$, so C is not bounded below and no reverse comparison exists.

The source check now shows that Bauerschmidt–Dagallier use the L^2 /stochastic-quantisation form, so their theorem cannot enter Theorem 3.2. Moreover, even a covariance-weighted estimate would not by itself validate the former Theorem 2.3 of `h6-attack.tex`; that proof has an independent divergent-trace error. The corrected conditional recursion and its additional hypotheses are in `remaining.tex`, Proposition 6.2.

5 Numerical verification

Both analytic inputs checked numerically ($m_0 = 1$; k_0 -integration on $[-4000, 4000]$ with $4 \cdot 10^6$ points; Gaussian mollifier $|\hat{\rho}(\varepsilon k_0)|^2 = e^{-(\varepsilon k_0)^2}$).

(a) Lemma 2.1				(b) Lemma 3.1, $k = 1$		
k	numeric	$1/(2\mu(k))$	rel. error	ε	$s_\varepsilon(1)$	$\leq \widehat{C}(1)?$
0	0.4999204	0.5000000	$1.6 \cdot 10^{-4}$	1.00	0.118866	yes
1	0.3534738	0.3535534	$2.2 \cdot 10^{-4}$	0.30	0.232172	yes
10	0.0496723	0.0497519	$1.6 \cdot 10^{-3}$	0.10	0.303518	yes
				0.03	0.337244	yes
				0.01	0.347981	yes
				0	0.353553	—

The residual error in (a) is the truncation of the $1/k_0^2$ tail at $|k_0| = 4000$, consistent in size with $2\mu(k)/(\pi \cdot 4000)$. In (b), s_ε increases monotonically to $\widehat{C}$ from below at every k tested, as Lemma 3.1 asserts.

6 Status

- **(T3) is proved** (Theorem 3.2), with no loss of constant, given (P-2d).
- **The BD premise fails:** [1] proves the unweighted form, not (P-2d). Hence its inequality cannot be passed through Theorem 3.2 to obtain (P-U).
- Tasks (T1), (T2), and (T4) are irrelevant to this route unless a genuinely covariance-weighted two-dimensional estimate is first established.

References

- [1] R. Bauerschmidt and B. Dagallier, *Log-Sobolev inequality for the φ_2^4 and φ_3^4 measures*, Comm. Pure Appl. Math. **77** (2024) 2579–2612; arXiv:2202.02295.
- [2] Milestone M3, `../m3/m3.tex`.